Practice workbook · Extraneous variables and how to control them

Participant and situational variables, ways of controlling them (standardisation, blind techniques) and the choice of experimental design.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-02

1. Question 1 · part b · 2 marks

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Worked-example retry · Attempt part b only.

1 Thomas recruited participants to investigate obedience to authority. The participants were asked to attend a university.

They were allocated to one of two conditions:

- Condition A: 30 participants were instructed to answer general knowledge questions in silence for 40 minutes by a man dressed in a uniform.
- Condition B: 30 participants were instructed to answer general knowledge questions in silence for 40 minutes by a man dressed in jeans and a t-shirt.

Thomas recorded how many participants obeyed the instruction to remain silent.

(a) Identify the dependent variable (DV) in Thomas' investigation.

(1)

- ☒ **A** Whether the man was wearing a uniform or jeans and a t-shirt.
- ☒ **B** The amount of time participants were given to answer the questions.
- ☒ **C** The number of participants who obeyed the man's instruction to remain silent.
- ☒ **D** The number of general knowledge questions participants were instructed to answer.

(b) Describe **one** participant variable that Thomas may have considered when conducting his investigation.

(2)

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1. Question 1 · part b · 2 marks

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Worked-example retry · Attempt part b only.

The results of Thomas' investigation are shown in **Table 1**.

	Obeyed instruction (remained silent)	Disobeyed instruction (did not remain silent)
Condition A: Man wearing uniform	22	8
Condition B: Man wearing jeans and t-shirt	11	19

Table 1

(c) Calculate the percentage of participants who **obeyed** the instruction, out of all the participants in the investigation.

(2)

Percentage .....

(d) Calculate the ratio of participants who **disobeyed** the instruction, to all participants taking part in the investigation.

(2)

Ratio .....

(e) Thomas concludes that the clothes worn by an authority figure influence levels of obedience.

Give **one** way Thomas' results could support this conclusion.

(1)

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(Total for Question 1 = 8 marks)

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1. Question 1 · part b · 2 marks · Working and answer

Working and answer

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2. Question 3 · part c · 2 marks

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Further practice · Attempt part c only.

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3 Becca wants to find out if there are sex differences in male and female brain functioning. She decides to use a laboratory experiment.

(a) State **one** feature of a 'laboratory experiment'.

(1)

(b) Describe how Becca could use volunteer sampling to gather the participants for her laboratory experiment.

(2)

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(c) Becca will give each participant a coordination task and a word-meaning task to test for sex differences in brain functioning. She decides to use a single-blind technique.

Describe **one** reason why Becca used a single-blind technique for her experiment.

(2)

(d) Becca conducted her laboratory experiment with 30 male and 25 female participants.

Calculate the ratio of male to female participants in Becca's sample.

(1)

Ratio .....

2. Question 3 · part c · 2 marks · Working and answer

Working and answer

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3. Question 4 · part d · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part d only.

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4 Ricky used a field experiment to investigate whether there is a difference between English and Sociology students' obedience to an authority figure.

He arranged for an English teacher and Sociology teacher to give their students an instruction to complete a piece of work. The teachers then left the classroom for 45 minutes.

Ricky recorded the number of students who did and did not complete the work. The results are shown in **Table 3**.

	Number of students who did complete the work	Number of students who did not complete the work
English students	16	8
Sociology students	15	10

Table 3

(a) Calculate the percentage of students who completed the work, out of all the students in the experiment.

(2)

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Percentage of students who completed the work .....

3. Question 4 · part d · 1 mark

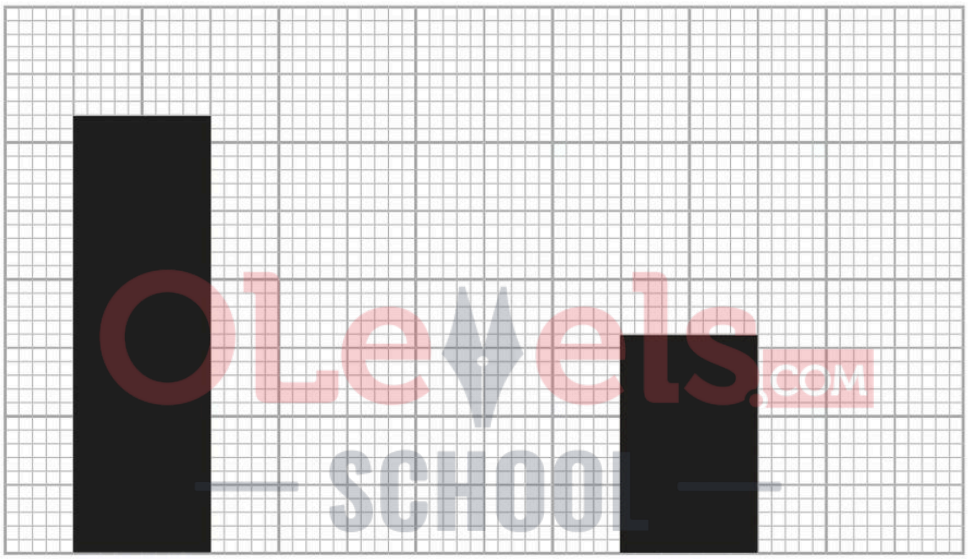
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Further practice · Attempt part d only.

(b) Complete the bar chart with the data from **Table 3** for the Sociology students, including the label and scale for the y-axis.

(3)

A bar chart to show the number of English and Sociology students who did and did not complete the work set by the teacher.



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3. Question 4 · part d · 1 mark

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Further practice · Attempt part d only.

(c) Ricky concluded that there was no significant difference between English and Sociology students' obedience to an authority figure.

Give **one** way that Ricky's findings could support this conclusion.

(1)

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(d) Give **one** participant variable that Ricky could have controlled for in his experiment.

(1)

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(Total for Question 4 = 7 marks)

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Working and answer

4. Question 1 · part b · 2 marks

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Further practice · Attempt part b only.

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1 Andrew conducted a laboratory experiment to investigate the capacity of short-term memory. He showed participants 20 trigrams and asked them to recall as many of the trigrams as they could.

(a) Andrew counted how many trigrams each participant recalled correctly.

Identify the type of data gathered by Andrew.

(1)

☒ **A** Qualitative data

☒ **B** Subjective data

☒ **C** Quantitative data

☒ **D** Secondary data

(b) Andrew used a standardised procedure when showing the trigrams to his participants.

Give **two** ways that Andrew could have standardised his procedure.

(2)

1 .....

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2 .....

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4. Question 1 · part b · 2 marks

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Further practice · Attempt part b only.

(c) Andrew sampled a total of 20 participants. A quarter of the participants were male.

Calculate the ratio of males to females in the participant sample used by Andrew in his experiment.

(2)

Ratio .....

(d) The number of trigrams recalled by participants correctly in Andrew's experiment are shown in **Table 1**.

	Number of trigrams recalled correctly				
	Between 1 and 4 trigrams	**Between 5 and 8 trigrams**	**Between 9 and 12 trigrams**	**Between 13 and 16 trigrams**	**Between 17 and 20 trigrams**
Males	0	2	3	0	0
Females	1	5	8	1	0

Table 1

Calculate the percentage of participants who recalled between 9 and 12 trigrams correctly, out of all the participants in the experiment.

(2)

Percentage .....

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P 7 0 4 8 6 A 0 4 4 8

4. Question 1 · part b · 2 marks · Working and answer

Working and answer

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5. Question 2 · part c · 2 marks

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Further practice · Attempt part c only.

2 Kamira is investigating morality. She wants to find out how likely people are to return lost items. Kamira conducted an observation for her investigation.

Kamira went to a large shopping centre and positioned herself in the food court where people often sit to eat and have been known to leave personal items behind when leaving.

When someone had left an item behind, Kamira observed the behaviour of all passers-by who saw the item. She recorded whether they:

- looked at the item, but then ignored it
- picked up the item and gave it to a member of staff
- walked away with the item.

(a) Describe why Kamira may have used an observation research method for her investigation.

(2)

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5. Question 2 · part c · 2 marks

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Further practice · Attempt part c only.

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(b) Draw a frequency table that Kamira could have used to record the data for her observation.

You must include appropriate column and row headings.

(3)

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(c) Describe **one** situational variable that Kamira may have considered when conducting her observation.

(2)

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(Total for Question 2 = 7 marks)

5. Question 2 · part c · 2 marks · Working and answer

Working and answer

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6. Question 2 · part b · 1 mark

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Further practice · Attempt part b only.

2 Jenna investigated whether stress affected performance on a memory test.

She recruited 10 participants from her university. Participants were asked to remember a list of 20 words. Fifteen minutes later, they were asked to write down as many words as they could remember. The same participants were tested twice:

- Test 1: one week before their examinations began.
- Test 2: one week after their examinations had finished.

(a) Describe how Jenna could use an opportunity sampling method to gather the participants for her investigation.

(2)

(b) State which experimental research design Jenna used in her investigation.

(1)

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6. Question 2 · part b · 1 mark

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Further practice · Attempt part b only.

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Jenna recorded the total number of correct words remembered by participants both before and after their examinations. Her results are shown in **Table 2**.

Participant	Total number of correct words remembered one week before examinations began (out of 20)	Total number of correct words remembered one week after examinations finished (out of 20)
A	5	8
B	6	7
C	7	9
D	4	8
E	5	7
F	3	7
G	9	9
H	10	11
I	6	6
J	5	8

Table 2

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(c) Calculate the mean score for the total number of correct words remembered one week **before** the participants' examinations began.

(1)

Mean .....

(d) Calculate the median score for the total number of correct words remembered one week **after** the participants' examinations finished.

(2)

Median .....

(Total for Question 2 = 6 marks)

6. Question 2 · part b · 1 mark · Working and answer

Working and answer

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